



A constrained intrinsic cohesive finite element method with little stiffness reduction for fracture simulation



Zhennan Zhang*, Yaxiong Chen

School of Naval Architecture, Ocean and Civil Engineering, Shanghai Jiao Tong University, Shanghai 200240, China

State Key Laboratory for GeoMechanics and Deep Underground Engineering, China University of Mining & Technology, Xuzhou 221008, China

ARTICLE INFO

Article history:

Received 18 June 2014

Received in revised form 24 December 2014

Accepted 29 January 2015

Available online 9 February 2015

Keywords:

Intrinsic cohesive finite element method

Constraint algorithm

Stiffness reduction

Dynamic fracture simulation

Extrinsic cohesive finite element method

ABSTRACT

To solve the stiffness reduction problem caused by cohesive surfaces in the intrinsic cohesive finite element model (CFEM), a series of constraint conditions are introduced into the algebraic equation system of the CFEM to constrain the separation of cohesive surfaces. To make the constrained cohesive surfaces active at the appropriate time, the constraint conditions will be removed in the high stress zone where fracture could initiate and propagate. The efficiency of the constrained CFEM, instead of being reduced by the constraint conditions, is significantly improved because the constrained cohesive surfaces are not counted in the numerical implementation.

© 2015 Elsevier Ltd. All rights reserved.

1. Introduction

The cohesive finite element model (CFEM) pioneered by Needleman and coworkers [1–3] is an efficient approach to simulate fracture. In this method, a set of cohesive surfaces is embedded along all element boundaries. The bulk element is characterized by a volumetric constitutive relation, while the cohesive surface element is characterized by a cohesive law. The fracture is allowed to initiate and propagate only along the prescribed cohesive surfaces. The fracture process occurs along arbitrary paths, so any pattern can be explicitly simulated without a separate fracture criterion and mesh modification. Due to these advantages, the CFEM has been extensively applied to dynamic fracture simulation.

However, a CFEM with an initially elastic cohesive law, also called an *intrinsic* CFEM, has a critical drawback that is the material stiffness is significantly reduced as the density of the cohesive surfaces increases. This leads to a competing requirement on the element size. On one hand, to accurately simulate the fields in the crack tip vicinity, the element size must be small enough. On the other hand, as the element size is refined, the cohesive surface-induced stiffness reduction will become increasingly significant, which will seriously affect the solution. So, another question arises: what element size should be adopted when using intrinsic CFEM? To answer this question, Tomar et al. [4] have suggested a lower and upper bound of the element size. If the element size is within the two bounds, the crack tip field can be simulated with reasonable accuracy and the influence of the induced stiffness reduction is acceptable. In addition to the element size, the stiffness reduction problem also can be remedied by setting an appropriate stiffness ratio of the cohesive surface to bulk element as suggested in [5–7]. Though these methods can relieve the stiffness reduction effect to a certain extent, the stiffness reduction problem of

* Corresponding author at: School of Naval Architecture, Ocean and Civil Engineering, Shanghai Jiao Tong University, Shanghai 200240, China. Tel./fax: +86 21 34204346.

E-mail address: zhennanzhang@sjtu.edu.cn (Z. Zhang).

Nomenclature

A	element area
c_R	Rayleigh wave speed
$\mathbf{C}$	damp matrix
$\tilde{\mathbf{C}}$	transformed damp matrix
d	node number at a spot
E	Young's modulus
$\mathbf{f}$	external force vector
$\tilde{\mathbf{f}}$	transformed external force vector
$\mathbf{F}$	deformation gradient
k_c	initial stiffness of cohesive surface
k_b	initial stiffness of bulk material
$\mathbf{K}$	stiffness matrix
$\tilde{\mathbf{K}}$	transformed stiffness matrix
m	dimension of problem
$\mathbf{M}$	mass matrix
$\tilde{\mathbf{M}}$	transformed mass matrix
$\mathbf{nip}$	constraint array
$\mathbf{ip}$	constraint array
p	restricted degrees of freedom
q	restricted degrees of freedom
$\mathbf{Q}$	constraint matrix
$\mathbf{x}$	material point vector in current configuration
$\bar{\mathbf{x}}$	material point vector in reference configuration
$\mathbf{s}$	nonsymmetric nominal stress tensor
S_{int}	internal cohesive surface area
S_{ext}	external surface area, respectively
T_n	normal traction applied on the cohesive surface
T_t	shear traction applied on the cohesive surface
$T_n^{\max}$	maximum traction of the cohesive law
$\mathbf{T}$	traction vector on a surface in the reference configuration
$\mathbf{u}$	displacement vector
V	volume
δ_n	scale length corresponding to the normal separation
δ_t	scale length corresponding to the shear separation
Δ	displacement jump across the cohesive surface
Δ_n	normal separation of cohesive surface
Δ_t	shear separation of cohesive surface
ϕ_n	normal separation work
ϕ_t	shear separation works
γ	relaxation coefficient
ν	Poisson's ratio
ρ	material density
$\boldsymbol{\sigma}$	stress tensor
$\dot{\epsilon}_{yy}$	strain rate

the intrinsic CFEM is not well addressed yet. To avoid the stiffness reduction problem, Camacho and Ortiz [8] and Ortiz and Pandolfi [9] inserted cohesive surfaces into the mesh where the stress exceeds the cohesive strength and developed the *extrinsic* CFEM. The extrinsic CFEM employs the initially stiff cohesive law, which has been extensively applied to fracture simulation, e.g. [10–13]. However, the extrinsic CFEM involves local mesh modification and node duplication, which reduces the computational efficiency. To remedy this drawback, Paulino et al. [14] and Zhang et al. [15] developed a topological data structure for extrinsic CFEM implementation.

Actually, during the implementation of the intrinsic CFEM, only a small percentage of the cohesive surfaces, which are located in the high stress zone, really function as designed. Most cohesive surfaces are 'idle'. These idle cohesive surfaces are not only unnecessary, but play a negative role by seriously reducing the material stiffness. If the separations of these idle cohesive surfaces are constrained by introducing a series of constraint conditions, the stiffness reduction problem of the intrinsic CFEM can be addressed to a large extent. In the high stress zone, where fracture can initiate and propagate, the constrained cohesive surfaces are relaxed to function as designed by removing the constraint conditions. This approach is the

basic idea of the present method. Because the constrained cohesive surfaces are not computed in the numerical implementation, the efficiency of the CFEM is significantly improved.

2. Problem formulation of CFEM

Let $\mathbf{x}$ denote a material point in the current configuration that is initially at $\bar{\mathbf{x}}$. Next, the displacement vector $\mathbf{u}$ and the deformation gradient $\mathbf{F}$ are, respectively, defined as $\mathbf{u} = \mathbf{x} - \bar{\mathbf{x}}$ and $\mathbf{F} = \partial \mathbf{x} / \partial \bar{\mathbf{x}}$. The principle of virtual work for CFEM can be written in the form [1,3]

$$\int_V \mathbf{s} : \delta \mathbf{F} dV - \int_{S_{int}} \mathbf{T} \cdot \delta \Delta dS = \int_{S_{ext}} \mathbf{T} \cdot \delta \mathbf{u} dS - \int_V \rho \frac{\partial^2 \mathbf{u}}{\partial t^2} \cdot \delta \mathbf{u} dV \quad (1)$$

where $\mathbf{s}$ is the nonsymmetric nominal stress tensor; Δ is the displacement jump across the cohesive surface; V , S_{int} and S_{ext} are the volume, internal cohesive surface area and external surface area, respectively; $\mathbf{T}$ is the traction vector on a surface in the reference configuration; and ρ is the material density.

The cohesive law is a traction–separation relationship, which intrinsically governs the fracture initiation and propagation. Many cohesive laws have been proposed, e.g., the Xu–Needleman [3], improved Xu–Needleman [16], bilinear [4], trapezoidal [17], irreversible [8,9] and hysteretic cohesive law [18].

3. Origin of the constrained intrinsic CFEM

The mechanism by which the cohesive surface induces a stiffness reduction can be analyzed using a specimen shown in Fig. 1; the specimen is a rectangular plate that is equally discretized into $N + 1$ quadrilateral elements and N cohesive surfaces with no thickness. The height of each quadrilateral element is $h = L / (N + 1)$ where L is the plate height. Let k_c and k_b denote the initial stiffness of the cohesive surface and bulk material, respectively. An infinitesimal traction, σ , is applied on the opposite sides of this plate. The total deformation by the bulk elements and cohesive surfaces is $\Delta^b = L \cdot \sigma / k_b$ and $\Delta^c = N \cdot \sigma / k_c$, respectively. The plate strain can be calculated using $\varepsilon = (\Delta^c + \Delta^b) / L = (N \cdot \sigma / k_c + L \cdot \sigma / k_b) / L$ and the apparent stiffness of plate is $k = \sigma / \varepsilon = k_b k_c / (k_b \cdot N / L + k_c)$. Thus, the relative stiffness of this plate can be expressed as

$$\tilde{k} = \frac{k}{k_b} = \frac{k_c / k_b}{N / L + k_c / k_b} \quad (2)$$

Eq. (2) suggests that increasing the cohesive surface density, N / L , can lead to a stiffness reduction. Another approach to reduce the stiffness reduction is to improve the stiffness ratio k_c / k_b . However, a stiffness ratio k_c / k_b that is too large can lead to spurious traction oscillations, which may cause erroneous crack patterns, as noted by De Borst et al. [19].

In this example, if we constrain the cohesive surface to have no separation by introducing a certain constraint condition rather than by improving the stiffness ratio k_c / k_b , the stiffness reduction problem can be solved. When the stress of the constrained cohesive surface exceeds a certain value that indicates fracture may initiate and propagate, the constrained cohesive surface will be relaxed to function as designed by removing the constraint condition. The present method is developed based on this philosophy.

4. Constraint algorithm for the intrinsic CFEM

4.1. Special case

To prevent the cohesive surface from separating, a pair of contact nodes of the cohesive element is coupled together so that they have the same displacement. This is performed by a constraint algorithm. To show this algorithm more clearly, we present an equation system with 4 independent equations and 4 unknowns as:

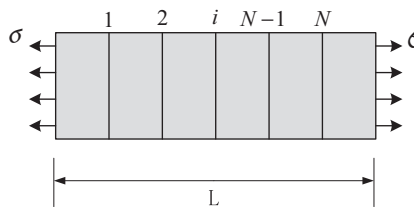


Fig. 1. Plate discretized into cohesive elements with N cohesive surfaces. (The cohesive surface has no thickness.)

$$\begin{bmatrix} k_{11} & k_{12} & k_{13} & k_{14} \\ k_{21} & k_{22} & k_{23} & k_{24} \\ k_{31} & k_{32} & k_{33} & k_{34} \\ k_{41} & k_{42} & k_{43} & k_{44} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix} \quad (\text{or } \mathbf{Ku} = \mathbf{f}) \quad (3)$$

where k_{ij} ($i, j = 1, \dots, 4$) are the coefficients; u_j ($j = 1, \dots, 4$) are the unknown variables; and f_i ($i = 1, \dots, 4$) are the known variables. If we impose a constraint condition $u_2 = u_3$ on Eq. (3), the solution of Eq. (3) can be obtained by solving the simultaneous equations

$$\begin{bmatrix} k_{11} & k_{12} + k_{13} & 0 & k_{14} \\ k_{21} + k_{31} & k_{22} + k_{23} + k_{32} + k_{33} & 0 & k_{24} + k_{34} \\ 0 & 0 & 0 & 0 \\ k_{41} & k_{42} + k_{43} & 0 & k_{44} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} f_1 \\ f_2 + f_3 \\ 0 \\ f_4 \end{bmatrix} \quad (4)$$

Note that Eq. (4) cannot be directly solved because its coefficient matrix is singular. However, the coefficient matrix of Eq. (4) is different from the usual singular matrix because the unknowns (u_1, u_2, u_4), but not u_3 , are uniquely determined by Eq. (4). Additionally, the components in the third column and third row are zero, which indicates that u_3 does not contribute to the product of $\mathbf{ku}$. Thus, Eq. (4) can be solved by an iterative method, e.g., the conjugate gradient method. Next, the solution of Eq. (3) with the constraint condition $u_3 = u_2$ can be obtained as (u_1, u_2, u_2, u_4) .

The transform from Eqs. (3) to (4) is

$$\mathbf{Q}^T \mathbf{K} \mathbf{Q} \mathbf{u} = \mathbf{Q}^T \mathbf{f} \quad (5)$$

where $\mathbf{Q}$ is the constraint matrix

$$\mathbf{Q} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (6)$$

According to the constraint condition and structural character of the constraint matrix, we find that the constraint matrix can be generated by directly operating the identity matrix. In this example, the constraint condition is $u_2 = u_3$, therefore the constraint matrix can be generated by moving the third column to the second column in the identity matrix. The solution with the constraint condition can be obtained by the transform $\mathbf{Qu}$ where $\mathbf{u} = [u_1, u_2, u_3, u_4]^T$.

4.2. General constraint algorithm

In the intrinsic CFEM, there are many cohesive surfaces. To constrain these cohesive surfaces, a series of constraint conditions should be imposed on the algebraic equations to ensure the nodes at a given spot have the same displacement. In other words, the nodes at a spot are coupled together by the constraint conditions. The spot here refers to a coordinate point where several nodes overlap in the meshing scheme of the CFEM, as shown in Fig. 2.

The nodes at a spot compose a coupled node set,

$$\{d_1, d_2, \dots, d_n\} \quad (7)$$

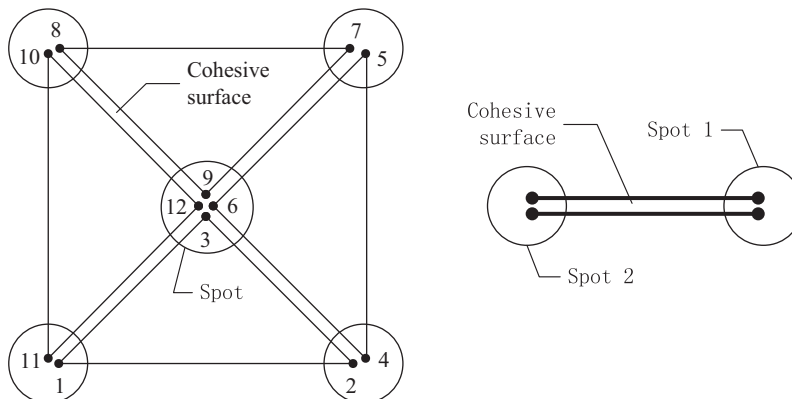


Fig. 2. Typical meshing scheme for CFEM where the nodes at a spot overlap each other.

where $d_1, d_2, \dots, d_n$ are the node numbers at this spot. For example, in Fig. 2, the nodes at the central spot compose a coupled node set $\{3, 6, 9, 12\}$.

A coupled node set can generate m coupled degrees-of-freedom (DOF) sets

$$\begin{aligned} & \{md_1, md_2, \dots, md_n\} \\ & \{md_1 - 1, md_2 - 1, \dots, md_n - 1\} \\ & \dots \\ & \{md_1 - m + 1, md_2 - m + 1, \dots, md_n - m + 1\} \end{aligned} \quad (8)$$

where $m = 1, 2, 3$ are for the 1D, 2D and 3D problems, respectively.

Let $\mathbf{u}$ stand for the displacement vector of the nodes. A coupled DOF set means that the components of $\mathbf{u}$ with the index of the set component are identical. For instance, the coupled DOF set $\{md_1, md_2, \dots, md_n\}$ means $u_{md_1} = u_{md_2} = \dots = u_{md_n}$. So, a coupled DOF set is actually a constraint condition.

For the large equation system of CFEM, two arrays are used to record the coupled DOF sets:

$$\begin{aligned} \mathbf{nip} &= [n_1, n_2, \dots, n_N]_{1 \times N} \\ \mathbf{ip} &= \begin{bmatrix} \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix}_{1 \times \sum n_i} \end{aligned} \quad (9)$$

where the array $\mathbf{nip}$ records the number of coupled DOFs in each set; $\mathbf{ip}$ records the coupled DOFs; N means the total number of coupled DOF sets; and n_i is the capacity of the i th coupled DOF set.

For the multiple coupled DOF sets, the key point is how to generate the constraint matrix to introduce the constraint conditions into the equation system. We will represent the generation rule of the constraint matrix by two coupled DOF sets. For two sets, $\{p_1, p_m\}$ and $\{q_1, q_i, q_n\}$, the generation of $\mathbf{Q}$ by an identity matrix can be illustrated as

$$\mathbf{Q} = \begin{matrix} & \cdots & p_1 & \cdots & p_m & \cdots & q_1 & \cdots & q_i & \cdots & q_n & \cdots \\ \vdots & & & & & & & & & & & \\ p_1 & \begin{bmatrix} 1 & \vdots & \cdots & \vdots & \cdots & \vdots & \cdots & \vdots & \cdots & \vdots & \cdots \\ \cdots & 1 & \cdots & 0 & \cdots & 0 & \cdots & 0 & \cdots & 0 & \cdots \\ \vdots & \vdots & \vdots & 1 & \vdots & \vdots & \cdots & \vdots & \cdots & \vdots & \vdots \\ p_m & 0 & 1 & \cdots & 0 & \cdots & 0 & \cdots & 0 & \cdots & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & 1 & \vdots & \cdots & \vdots & \cdots & \vdots & \vdots \\ q_1 & 0 & 0 & \cdots & 0 & \cdots & 1 & \cdots & 0 & \cdots & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & 1 & \vdots & \cdots & \vdots & \vdots \\ q_i & 0 & 0 & \cdots & 0 & \cdots & 1 & \cdots & 0 & \cdots & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & 1 & \vdots & \vdots \\ q_n & 0 & 0 & \cdots & 0 & \cdots & 1 & \cdots & 0 & \cdots & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & 1 \end{bmatrix} & \cdots \end{matrix} \quad (10)$$

where M is the number of equations.

As indicated by Eq. (10), to generate the constraint matrix, an identity matrix with dimension $neq \times neq$ should be defined beforehand where neq is the total number of equations. Next, each coupled DOF set should be identified by $\mathbf{nip}$ and $\mathbf{ip}$. For each coupled DOF set, the columns corresponding to the other components of this set in the pre-defined identity matrix are moved to the column corresponding to the first component of this set.

The usual governing equation for the dynamic problem is

$$\mathbf{M}\ddot{\mathbf{u}} + \mathbf{C}\dot{\mathbf{u}} + \mathbf{K}\mathbf{u} = \mathbf{f} \quad (11)$$

where $\mathbf{M}$ is the mass matrix; and $\mathbf{C}$ the damping matrix.

In the present method, the nodes at a spot are initially coupled together and will be decoupled at the high stress zone. The node velocity and acceleration should comply with the same constraint conditions as the node displacement. Therefore, the governing equation with constraint conditions has the following form:

$$\mathbf{Q}^T \mathbf{M} \ddot{\mathbf{u}} + \mathbf{Q}^T \mathbf{C} \dot{\mathbf{u}} + \mathbf{Q}^T \mathbf{K} \mathbf{u} = \mathbf{Q}^T \mathbf{f} \text{ (or } \tilde{\mathbf{M}} \ddot{\mathbf{u}} + \tilde{\mathbf{C}} \dot{\mathbf{u}} + \tilde{\mathbf{K}} \mathbf{u} = \tilde{\mathbf{f}}) \quad (12)$$

4.3. Verification example for the constraint algorithm

To verify the constraint algorithm, we will simulate a rectangular plate (Fig. 3a) that is subjected to a uniaxial tensile stress. The meshing scheme and boundary conditions are shown in Fig. 3b. The nodes {3, 6, 9, 12} are located at the plate center. A displacement $u_y = 1.0 \times 10^{-4}$ m is applied to node 7 and 8. The Young's modulus of the bulk material is $E = 10.0$ GPa and the Poisson's ratio is $\nu = 0.2$.

The coupled node sets are {1, 11}, {2, 4}, {3, 6, 9, 12}, {5, 7} and {8, 10}. The corresponding constraint arrays are

$$\begin{aligned} \mathbf{nip} &= [2, 2, 2, 2, 4, 4, 2, 2, 2, 2] \\ \mathbf{ip} &= [2, 22, 1, 21, 4, 8, 3, 7, 6, 12, 18, 24, 5, 11, 17, 23, 10, 14, 9, 13, 16, 20, 15, 19] \end{aligned} \quad (13)$$

The conjugate gradient method is used to solve the equations. The simulation results are listed in Table 1. As seen in Table 1, the nodes in a coupled node set have the same displacement, and the node force at a spot is balanced. The simulated results are consistent with the elastic solutions. This result suggests that the cohesive surfaces are successfully constrained and the constraint algorithm is effective.

4.4. Relaxation of constrained cohesive surface

Initially, all cohesive surfaces are restricted by the constraint conditions. As the load increases, the cohesive surfaces in the high stress zone, where the crack can initiate and propagate, should be relaxed so that they can be active at the appropriate time. Because a cohesive surface involves two spots, as shown in Fig. 2, the relaxation of the cohesive surface can be achieved by decoupling the nodes at these spots. The high stress zone is indicated by the spot stress. So, if the stress at a certain spot exceeds a critical value, the nodes at this spot and the corresponding coupled DOF sets will be decoupled. Because each spot is related to several bulk elements, the spot stress can be computed by weighting the stresses of the related bulk elements with respect to element volume as follows

$$\boldsymbol{\sigma} = \frac{\sum A_i \boldsymbol{\sigma}_i}{\sum A_i} \quad (14)$$

where $\boldsymbol{\sigma}$ is the spot stress tensor; A_i is the volume of the i th bulk element related to a given spot; and $\boldsymbol{\sigma}_i$ the stress tensor of the i th element.

To ensure that the cohesive surface is activated in time, we relax the cohesive surface before the cohesive stress reaches the cohesive strength. But, on the other hand, the cohesive surface should not be relaxed much 'earlier' to ensure that the constraint condition can work to reduce the stiffness reduction. Otherwise, the constraint algorithm does not make sense. Because the constrained cohesive surface has no separation, its cohesive stress cannot be directly computed. The maximum principal stress of the spot stress tensor is a measure of a constrained cohesive surface. So, the decoupling criterion for the nodes at a spot can be written as

$$\sigma_1 > \gamma T_n^{\max} \quad (15)$$

where σ_1 is the maximum principal stress of the spot stress tensor; $T_n^{\max}$ is the maximum traction of the cohesive law; and γ is the relaxation coefficient, $0 < \gamma < 1$, whose effect on the simulation results will be discussed in Section 6.

Once a spot is decoupled, all cohesive surfaces related to this spot are relaxed. The transition of a cohesive surface from the constrained to the relaxed state is shown in Fig. 4, where a separation jump from zero to the separation specified by the

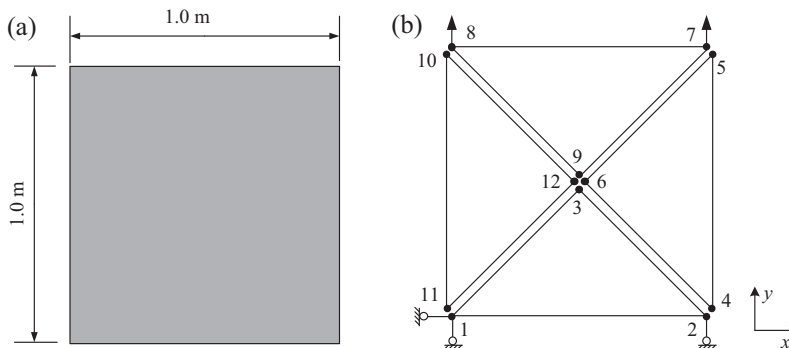
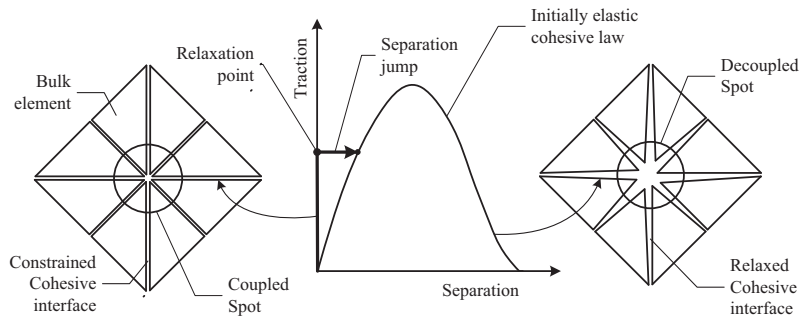


Fig. 3. Verification example for the constraint algorithm: (a) dimensions of the specimen and (b) meshing scheme and boundary conditions.

Table 1Simulated node displacement in the x-coordinate, u_x , and y-coordinate, u_y , (Unit: m) and the corresponding node force, f_x , f_y (Unit: N).

Node	1	2	3	4	5	6	7	8	9	10	11	12
$u_x \times 10^{-4}$	0.0	-0.2	-0.1	-0.2	-0.2	-0.1	-0.2	0.0	-0.1	0.0	0.0	-0.1
$u_y \times 10^{-4}$	0.0	0.0	0.5	0.0	1.0	0.5	1.0	1.0	0.5	1.0	0.0	0.5
$f_x \times 10^5$	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
$f_y \times 10^5$	-2.5	-2.5	5.0	-2.5	2.5	0.0	2.5	2.5	-5.0	2.5	-2.5	0.0

**Fig. 4.** Decoupling process of the spot and transition process of the cohesive surface from the constrained to the relaxed state.

traction–separation law occurs. The relaxed cohesive surface will be governed by the initial elastic cohesive law. This is similar yet different from the extrinsic CFEM. In the extrinsic CFEM, the cohesive surface is inserted into the mesh and is governed by the initially stiff cohesive law when the cohesive stress reaches the cohesive strength. Therefore, no separation jump occurs when inserting a cohesive surface. Another difference between the constrained and extrinsic CFEM is the cohesive surfaces previously existed in the mesh for the constrained CFEM, while they are inserted in the extrinsic CFEM. Therefore, the constrained CFEM does not involve local mesh modification and node duplication. The constrained CFEM inherits the characteristics of both the intrinsic and extrinsic CFEM.

4.5. Numerical implementation aspects

For numerical implementation of the present method, the following points should be clarified:

The constraint arrays **nip** and **ip** should first be established according to the meshing scheme. The establishment method has been described in Section 4.3. At the beginning of the numerical implementation, all cohesive surfaces are constrained. The constrained cohesive surfaces will not be computed in implementation because they do not work. The linkage between the bulk elements are fulfilled by the constraint conditions. Each cohesive surface involves two spots. As long as one of spots is decoupled, the cohesive surface is relaxed and will be computed.

Compared with the usual CFEM, the additional operations in the present method are the matrix–vector products $\mathbf{Q}\mathbf{u}$ and $\mathbf{Q}^T\mathbf{f}$. To compute them, it is unnecessary to explicitly generate the constraint matrix $\mathbf{Q}$, but directly operate on the vector based on **nip** and **ip** according to the structural character of the constraint matrix. The implicit time integration scheme involves solving the equation system $\mathbf{Q}^T\mathbf{K}\mathbf{Q}\mathbf{u} = \mathbf{Q}^T\mathbf{f}$ using the iterative computation method, e.g., the conjugate gradient method, which is the preferable option. The central operator is the product $\mathbf{Q}^T\mathbf{K}\mathbf{Q}\mathbf{u}$, which can be computed as $\mathbf{Q}^T\mathbf{K}\mathbf{Q}\mathbf{u} = \mathbf{Q}^T(\mathbf{K}(\mathbf{Q}\mathbf{u}))$. Using this computation scheme, we do not need to make any change to the storage format of the global stiffness matrix $\mathbf{K}$. This is quite important for finite element routines. When the vector $\mathbf{u}$ meets the convergence condition after many iterations, the solution with the constraint conditions is obtained by $\mathbf{Q}\mathbf{u}$.

After each time step, update the status (coupled or decoupled) of spots. If the spot stress meets the criterion of Eq. (15), all the coupled DOF sets related to this spot are decoupled. The decoupling process can be achieved by either skipping the corresponding segmental operation on the vector $\mathbf{u}$ in the operation of matrix–vector product $\mathbf{Q}\mathbf{u}$ or removing the corresponding coupled DOF set in **nip** and **ip**. The relaxation process is irreversible.

5. Efficiency check

Because a series of constraint conditions are introduced in solving equations, a major concern is how these constraint conditions affect the speed. To check this problem, we perform a comparative analysis on the efficiency. In this analysis, we will use the usual CFEM and the constrained CFEM to simulate a square plate (Fig. 5a) subjected to a uniaxial tensile stress. The cross-triangular meshing scheme is adopted, which is composed of square quadrilaterals with dimensions of

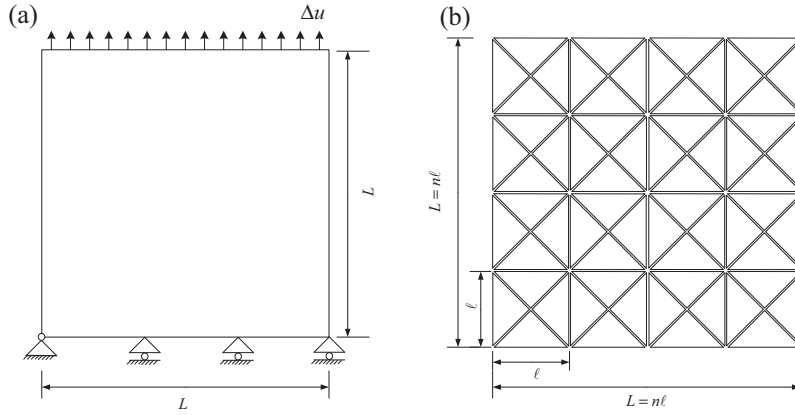


Fig. 5. Simulation specimen for efficiency check: (a) boundary conditions; and (b) meshing scheme.

$\ell \times \ell = 12.5 \mu\text{m} \times 12.5 \mu\text{m}$ (Fig. 5b). The quadrilateral size is fixed. The dimensions of the specimen L vary with the element number. The bulk material properties are $E = 3.24 \text{ GPa}$ and Poisson's ratio $\nu = 0.35$.

The following cohesive law [16] is employed

$$\begin{aligned} T_n &= \frac{\phi_n}{\delta_n^2} \cdot \Delta_n \cdot \exp\left(-\frac{\Delta_n}{\delta_n}\right) \cdot \exp\left(-\frac{\Delta_t^2}{\delta_t^2}\right) \\ T_t &= \frac{2\phi_t}{\delta_t^2} \cdot \Delta_t \cdot \exp\left(-\frac{\Delta_n}{\delta_n}\right) \left(1 + \frac{\Delta_n}{\delta_n}\right) \exp\left(-\frac{\Delta_t^2}{\delta_t^2}\right) \end{aligned} \quad (16)$$

where T_n , T_t are the normal and shear traction applied on the cohesive surface, respectively; ϕ_n , ϕ_t are, respectively, the normal and shear separation works; δ_n , δ_t are the scale length corresponding to the normal and shear separation, respectively; Δ_n , Δ_t are, respectively, the normal and shear separation of the cohesive surface. In this example, the parameters: $\phi_n = 330.0 \text{ J/m}^2$, $\phi_t = 330.0 \text{ J/m}^2$ and $\delta_n = \delta_t = 4.0 \times 10^{-7} \text{ m}$. A displacement-controlled loading scheme is adopted. The simulation is conducted on a personal computer with 2 GB internal storage and an I3-2120 CPU @ 3.3 GHz.

5.1. Effect of DOFs

To verify the effect of the DOFs, we change the element number in different cases. In each case, the load step is $\Delta u = 0.1\varepsilon_b L$ where the strain ε_b is defined as $\varepsilon_b = 0.1\delta_n/\ell = 3.2 \times 10^{-3}$. Based on this loading scheme, the separation of the cohesive surface is approximately invariant as the element number increases. We account for the time consumed for one load step in every case. The simulated result is shown in Fig. 6a, which indicates that the usual CFEM consumes much more time than the constrained CFEM. With increasing DOFs, the efficiency of the constrained CFEM becomes increasingly significant compared to the usual CFEM.

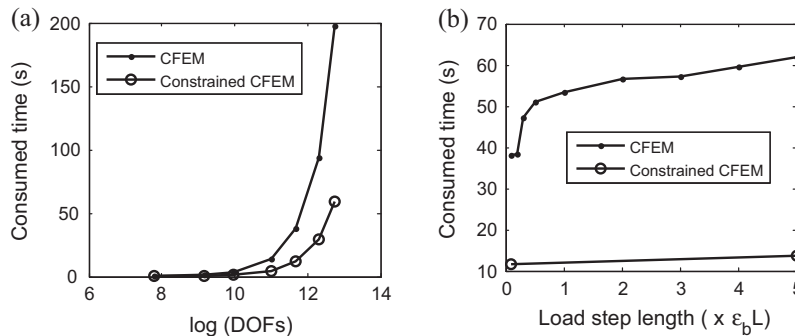


Fig. 6. Efficiency check results: (a) effect of the DOFs; and (b) effect of the load step length.

5.2. Load step effect

To check the effect of the load step length, we change the load step length from $\Delta u = 0.1 \varepsilon_b L$ to $\Delta u = 5.0 \varepsilon_b L$ in the meshing scheme of $L = 70\ell$ (117,600 DOFs). The time consumed for one load step is shown in Fig. 6b, which demonstrates that the time consumed in the usual CFEM increases with the load step length, while it is almost invariant in the constrained CFEM. This suggests that the constrained CFEM is insensitive to the load step length, which is very important for large scale computations.

The efficiency analysis above is based on the fact that all the cohesive surfaces are constrained in the constrained CFEM. However, in the implementation of the constrained CFEM, only a small percentage of the cohesive surfaces are relaxed. Therefore, the effect of the relaxed cohesive surfaces on the efficiency is limited.

6. Simulation example

To examine the performance of the present method in a dynamic fracture simulation, we simulate an example reported in [20]. The dimensions of the specimen are a 3 mm square block with an edge crack extending 0.25 mm into the block from the left edge (Fig. 7). The block is loaded uniaxially by constantly increasing the relative y -displacements of the top and bottom boundaries with a strain rate of $\dot{\varepsilon}_{yy} = 2000.0 \text{ s}^{-1}$. The side boundaries are stress free. The meshing scheme shown in Fig. 5b is adopted, which is composed of 120×120 25 μm quadrilaterals. The parameters in the present simulation are $E = 16.2 \text{ GPa}$, $\nu = 0.35$ and material density $\rho = 1190 \text{ kg/m}^3$. The Rayleigh wave speed is approximately $c_R = 2097 \text{ m/s}$. The cohesive law of Eq. (16) is adopted, whose parameters are $\phi_n = 330.0 \text{ J/m}^2$, $\phi_t = 330.0 \text{ J/m}^2$, $\delta_n = \delta_t = 4.0 \times 10^{-7} \text{ m}$, which leads to $T_{\max} = 303.0 \text{ MPa}$. To solve the dynamic equations, the explicit time-integration scheme is adopted with a time interval of $\Delta t = 0.1 \times 10^{-9} \text{ s}$.

To track the fracture path, we approximately take the array of the over-separated cohesive surfaces as the fracture trajectory. Here, the over-separated cohesive surface is the one whose separation meets

$$\sqrt{\Delta_n^2 + \Delta_t^2} \geq 10\delta_n \quad (17)$$

According to the cohesive law (Eq. (16)), when separation of the cohesive surface satisfies the condition in Eq. (17), the cohesive traction is small enough to be negligible. Hence, it is reasonable to use Eq. (17) as the over-separation condition. To represent the irreversible property of fracture, the over-separated cohesive surface will no longer have a tensile strength even though its separation recovers to zero.

The simulated dynamic fracture trajectory is shown in Fig. 8, which demonstrates that the fracture with a lower relaxation coefficient, e.g., $\gamma = 0.5$, is more likely to branch than a higher one coefficient, e.g., $\gamma = 0.9$. The simulated fracture morphologies are similar to those reported in [21]. In both cases of these two relaxation coefficients, the typical feature of instable fracture, i.e., the ‘mirror–mist–hackle–branching’ phenomenon observed in [22–25], is generally reproduced, as shown in Fig. 9.

During fracture propagation, the coupled spots are relaxed surround the crack tip. The percentage of the relaxed spots is indicated in Fig. 10. The higher the relaxation coefficient is, the smaller the percentage of relaxed spots and the less the stiffness reduction. Even for the low relaxation coefficient case ($\lambda = 0.5$), a majority of the cohesive surfaces are constrained

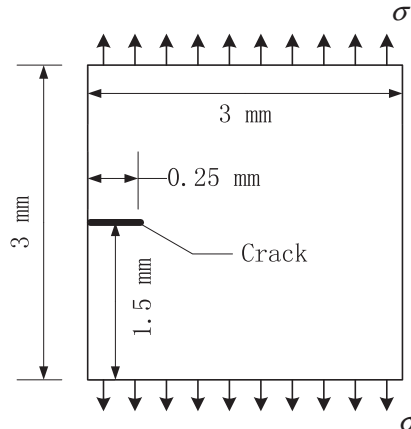


Fig. 7. Simulation specimen for dynamic fracture.

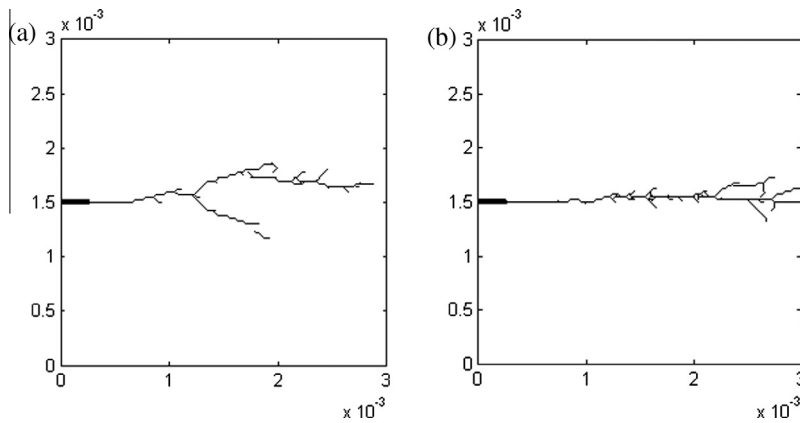


Fig. 8. Simulated dynamic crack trajectory for (a) $\gamma = 0.5$ and (b) $\gamma = 0.9$.

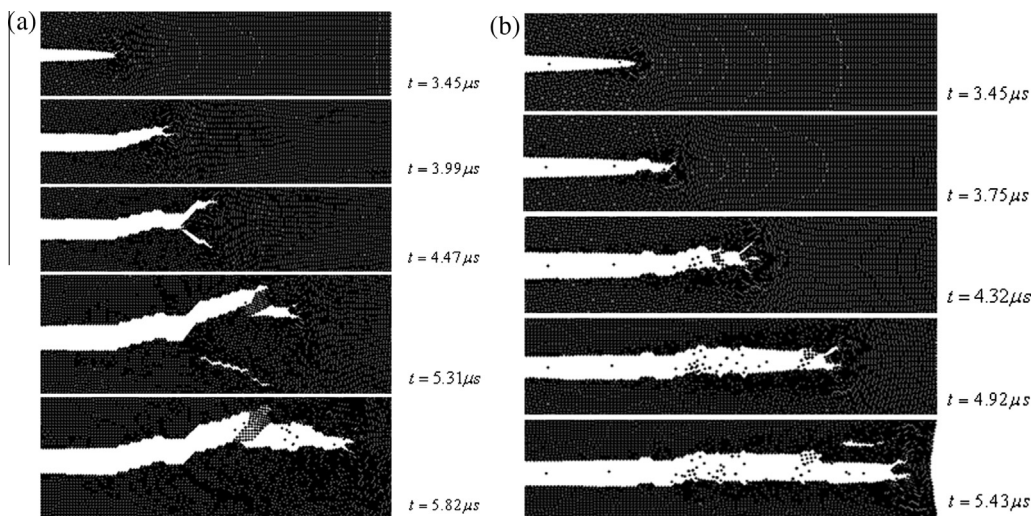


Fig. 9. Fracture propagation process for (a) $\gamma = 0.5$ and (b) $\gamma = 0.9$.

all the time. Fig. 10 also demonstrates that the relaxed spots envelop the fracture path, which ensures that the cohesive surfaces can be activated at the appropriate time in the fracture zone.

Fig. 11 shows the crack speeds before instability for the cases of $\gamma = 0.9$ and $\gamma = 0.5$. In Fig. 11, it is seen that the two crack speeds are almost the same though the time to fracture with $\gamma = 0.9$ is earlier than $\gamma = 0.5$. The simulated mean crack speed before instability is about $900 \text{ m/s} \approx 0.43c_R$. According to [25–27], the crack speed range for instability onset is approximately $0.4c_R \sim 0.57c_R$. The simulated crack speed before instability just falls into this scope.

Fig. 12 shows the applied traction on the specimen vs. the applied time. With reference to Fig. 9, it is found that the crack still propagates in a stable manner even when the applied traction reaches its peak value. The relaxation coefficient has little influence on the reaction of the specimen according to Fig. 12.

The present method provides an approach to addressing the stiffness reduction problem of the intrinsic CFEM; therefore, it can be extensively applied to fracture simulation. In the fracture problem of materials with micro- or nano-structures, a cohesive law is frequently used to describe the interface actions. For example, Shao et al. [28] used the triangular and trapezoidal cohesive laws to describe the interaction between the bridging platelet and organic phase of nacre. If this micro-structure is modeled with the cohesive finite element method, the present method can be extended to micro fracture cases because the constrained cohesive surface can describe the behaviors of the interface before debonding.

The present method also can be extended to the extrinsic CFEM if the relaxation coefficient is $\gamma = 1.0$, which means that the cohesive surface is relaxed when the cohesive stress reaches the cohesive strength, and the initially stiff cohesive law is adopted, as shown in Fig. 13. If the extrinsic CFEM takes the present constraint method, local mesh modification and node duplication are avoided.

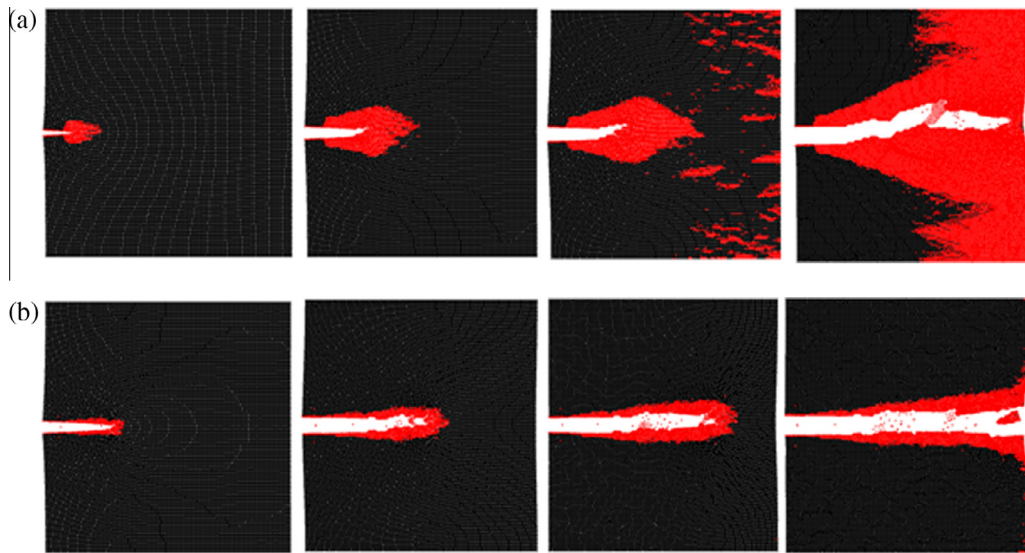


Fig. 10. Decoupled spots (highlighted by the red dot) vs. the coupled spots (highlighted by the black dots) during fracture propagation for (a) $\gamma = 0.5$ and (b) $\gamma = 0.9$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

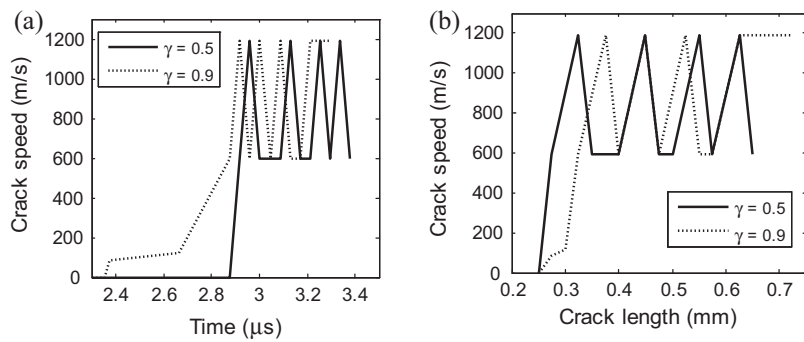


Fig. 11. Crack speed before instability: (a) crack speed vs. time and (b) crack speed vs. crack length.

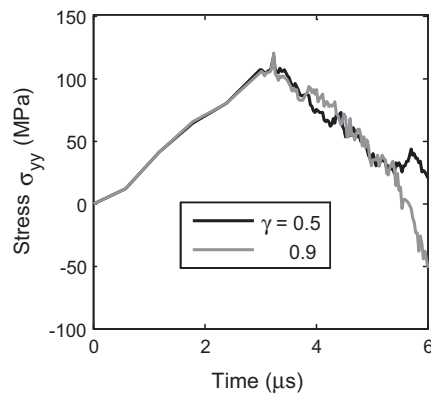


Fig. 12. Applied traction vs. time.

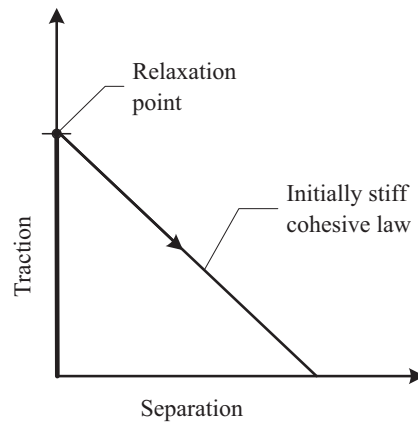


Fig. 13. Extension of the constraint method to extrinsic CFEM.

7. Conclusions

The stiffness reduction induced by cohesive surfaces has been a weakness of the intrinsic CFEM. By introducing a series of constraint conditions to the algebraic equations of CFEM, the cohesive surfaces for low stress zones where no fracture can initiate are deactivated. Using this method, the stiffness reduction problem is addressed. Moreover, the efficiency of the CFEM is also improved because the deactivated cohesive surfaces, which are the majority of the surfaces, are not used in the numerical implementation. The simulation results suggest that the present method can capture the characteristics of dynamic fracture by introducing a series of constraint conditions.

Not only is the constraint method applicable to the intrinsic CFEM, but also applicable to the extrinsic CFEM. With the constraint algorithm, the extrinsic CFEM can avoid the local mesh modification and the node duplication.

The relaxation coefficient should take a moderate value. If it is too high, the node oscillation due to the relaxed separation jump may have non-negligible effect while if too low, the computation efficiency would be significantly reduced and the stiffness reduction would become significant due to many unnecessarily-relaxed cohesive surfaces. So, the suggested range of relaxation coefficient is from 0.5 to 0.9 according to the simulation examples.

Acknowledgments

The present work is supported by the National Natural Science Foundation of China (No. 11172172), the National Basic Research Program of China (Grant No. 2011CB013505) and the State Key Laboratory for GeoMechanics and Deep Underground Engineering, China University of Mining & Technology (Grant No. SKLGDUEK1203). They are gratefully acknowledged.

Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.engfracmech.2015.01.030>.

References

- [1] Needleman A. A continuum model for void nucleation by inclusion debonding. *J Appl Mech* 1987;54:525–31.
- [2] Needleman A, Tvergaard V. An analysis of ductile rupture modes at crack tip. *J Mech Phys Solids* 1987;35:151–83.
- [3] Xu XP, Needleman A. Numerical simulations of fast crack growth in brittle solids. *J Mech Phys Solids* 1994;42(9):1397–434.
- [4] Tomar V, Zhai J, Zhou M. Bounds for element size in a variable stiffness cohesive finite element model. *Int J Numer Methods Engng* 2004;61:1894–920.
- [5] Zou Z, Reid SR, Li S, Soden PD. Modelling interlaminar and intralaminar damage in filament wound pipes under quasi-static indentation. *J Compos Mater* 2002;36:477–99.
- [6] Camanho PP, DaVila CG, de Moura MF. Numerical simulation of mixed-mode progressive delamination in composite materials. *J Compos Mater* 2003;37(16):1415–38.
- [7] Turon A, Davila CG, Camanho PP, Costa J. An engineering solution for mesh size effects in the simulation of delamination using cohesive zone models. *Engng Fract Mech* 2007;74:1665–82.
- [8] Camacho GT, Ortiz M. Computational modeling of impact damage in brittle materials. *Int J Solids Struct* 1996;33:2899–938.
- [9] Ortiz M, Pandolfi A. Finite deformation irreversible cohesive elements for three dimensional crack-propagation analysis. *Int J Numer Methods Engng* 1999;44(9):1267–82.
- [10] Pandolfi A, Ortiz M. An efficient adaptive procedure for three-dimensional fragmentation simulations. *Engng Comput* 2002;18:148–59.
- [11] Yu C, Pandolfi A, Ortiz M, Coker D, Rosakis AJ. Three-dimensional modeling of intersonic shear-crack growth in asymmetrically loaded unidirectional composite plates. *Int J Solids Struct* 2002;39:6135–57.
- [12] Elices M, Rocco C, Rosello C. Cohesive crack modelling of a simple concrete: experimental and numerical results. *Engng Fract Mech* 2009;76:1398–410.

- [13] Pandolfi A, Yu C, Corigliano A, Ortiz M. Modeling dynamic fracture in transversely isotropic composites: a cohesive approach. California Institute of Technology; 2000. Caltech (ASCI/2000.097).
- [14] Paulino GH, Celes W, Espinha R, Zhang Z. A general topology-based framework for adaptive insertion of cohesive elements in finite element meshes. *Engng Comput* 2008;24:59–78.
- [15] Zhang Z, Paulino GH, Celes W. Extrinsic cohesive modeling of dynamic fracture and microbranching instability in brittle materials. *Int J Numer Methods Engng* 2007;72:893–923.
- [16] Van den Bosch M, Schreurs P, Geers M. An improved description of the exponential Xu and Needleman cohesive zone law for mixed-mode decohesion. *Engng Fract Mech* 2006;73(9):1220–34.
- [17] Cornec A, Scheider I, Schwalbe K. On the practical application of the cohesive model. *Engng Fract Mech* 2003;70:1963–87.
- [18] Serebrinsky S, Ortiz M. A hysteretic cohesive-law model of fatigue-crack nucleation. *Scripta Mater* 2005;53:1193–6.
- [19] De Borst R, Remmers JJC, Needleman A. Mesh-independent discrete numerical representations of cohesive-zone models. *Engng Fract Mech* 2006;73:160–77.
- [20] Falk ML, Needleman A, Rice JR. A critical evaluation of dynamic fracture simulations using cohesive surfaces. *J Phys IV* 2001;11:43–52.
- [21] Klein PA, Foulk JW, Chen EP, Wimmer SA, Gao HJ. Physics-based modeling of brittle fracture: cohesive formulations and the application of meshfree methods. *Theor Appl Fract Mech* 2001;37:99–166.
- [22] Cramer T, Wanner A, Gumbsch P. Energy dissipation and path instabilities in dynamic fracture of silicon single crystals. *Phys Rev Lett* 2000;85:788–91.
- [23] Fineberg J, Gross SP, Marder M, Swinney HL. Instability in dynamic fracture. *Phys Rev Lett* 1991;67:457–60.
- [24] Ravi-Chandar K, Knauss WG. An experimental investigation into dynamic fracture: I. Crack initiation and arrest. *Int J Fract* 1984;25:247–62.
- [25] Abraham FF, Brodbeck D, Rafey RA, Rudge WE. Instability dynamics of fracture: a computer simulation investigation. *Phys Rev Lett* 1994;73:272–5.
- [26] Abraham FF, Brodbeck D, Rudge WE, Xu XP. A molecular dynamics investigation of rapid fracture mechanics. *J Mech Phys Solids* 1997;45:1595–619.
- [27] Sharon E, Fineberg J. Confirming the continuum theory of dynamic brittle fracture for fast cracks. *Nature* 1999;397:333–5.
- [28] Shao Y, Zhao HP, Feng XQ, Gao HJ. Discontinuous crack-bridging model for fracture toughness analysis of nacre. *J Mech Phys Solids* 2012;60:1400–19.